Greedy Algorithm

Tuesday, August 25, 2026 10:10 PM

① Dijkstra
② kruskal/ Prims } greedy strategy/approach
③ Interval Scheduling problem
④ Huffman Encoding

Optimization problem
min max

~~①~~ DP S.C↑
~~②~~ Greedy
~~③~~ Bruteforce T.C↑

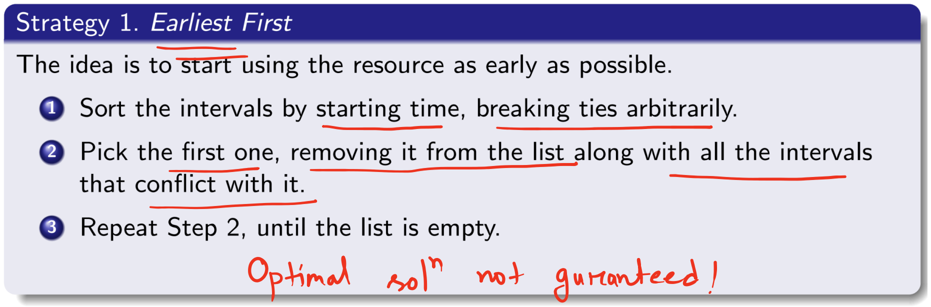
Greedy Strategy

→ Current context এ best possible decision নাও।

→ Two characteristics:

① Greedy choice property

② Optimal substructure



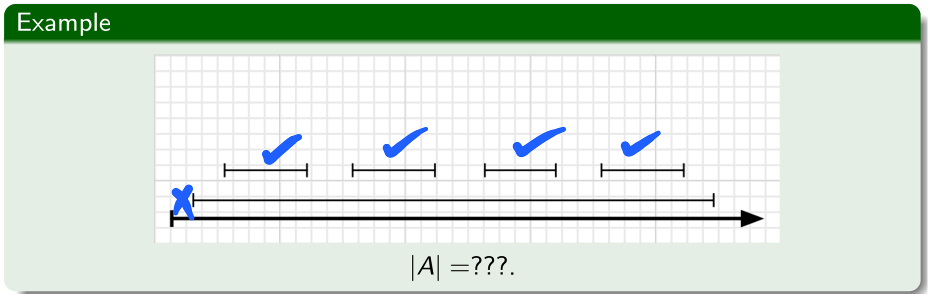
Strategy 1. *Earliest First*

The idea is to start using the resource as early as possible.

1. Sort the intervals by starting time, breaking ties arbitrarily.
2. Pick the first one, removing it from the list along with all the intervals that conflict with it.
3. Repeat Step 2, until the list is empty.

Optimal solⁿ not guranteed!

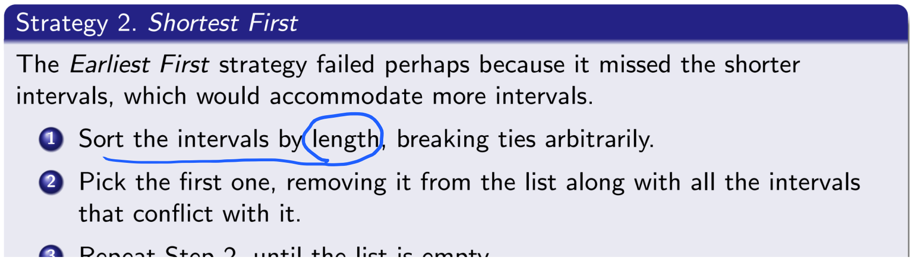
A = {5}
|A| = 1



Example

$|A| = ???.$

5



Strategy 2. *Shortest First*

The *Earliest First* strategy failed perhaps because it missed the shorter intervals, which would accommodate more intervals.

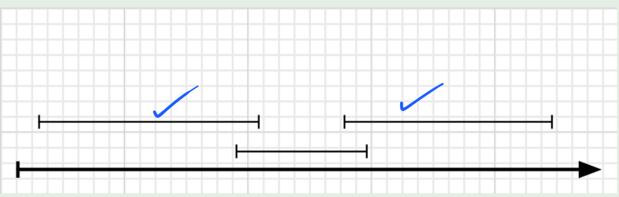
1. Sort the intervals by length, breaking ties arbitrarily.
2. Pick the first one, removing it from the list along with all the intervals that conflict with it.
3. Repeat Step 2, until the list is empty.

❸ Repeat Step 2, until the list is empty.

Optimal sol^n not guranteed!

Example

A = {3} ✓



|A| =???.

✓ 3m → 3

5m → 1

5m → 2

Strategy 3. *Least-conflict First*

The *Shortest First* strategy failed perhaps because the shorter ones had more conflicts, and ruled out too many intervals in the process.

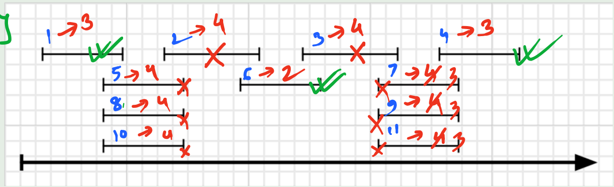
1. ✓ Sort the intervals by the number of other intervals which conflict with it.
2. Pick the first one, removing it from the list along with all the intervals that conflict with it.
3. Repeat Step 2, until the list is empty.

Optimal sol^n not guranteed!

Example

A = {1, 6, 4}

|A| = 3



|A| =???.

6 ✓

1 ✓

4 ✓

~~2~~

~~3~~

~~5~~

~~8~~

~~10~~

7

9

11

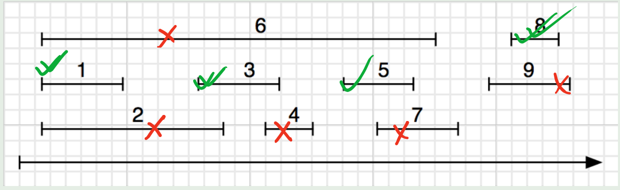
Strategy 4. *Finish First*

The idea is to free up the resource as early as possible.

1. Sort the intervals by the finishing time, breaking ties arbitrarily.
2. Pick the first one, removing it from the list along with all the intervals that conflict with it.
3. Repeat Step 2, until the list is empty.

Optimal sol^n is guranteed!

Example



|A| =???.

~~1~~ ✓

~~2~~

~~3~~ ✓

~~4~~

5 ✓

~~6~~

~~7~~ ✗

8 ✓

~~9~~

Extension: weighted interval scheduling problem

Definition (Weighted interval scheduling problem)

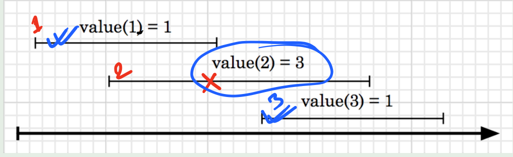
Given a set of schedules $I = \{I_i\}$, with associated weights $W = \{w_i\}$, find $A \subseteq I$ such that the members of A are non-conflicting *and* the total weight $\sum_{i \in A} w_i$ is maximized.

1 → 1

~~2 → 3~~

3 → 1

Example



$|A| = ???$, $\sum_{i \in A} w_i = ???$.

$A = \{1, 3\}$

$W_T = 1 + 1 = 2$

Greedy can not gurantee optimal sol^n

Huffman Encoding → bit compression technique / strategy

50 GB → 35 GB

compression

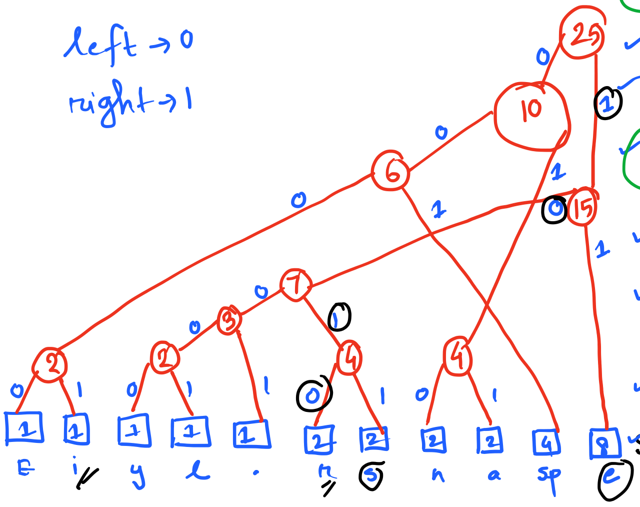
- Consider the following short text:

Eerie eyes seen near lake.

left → 0

right → 1

Character	Count	Huffman Code	No. of bits	Total Bit
E	1	0000	4	1×4 = 4
e	8	11	2	8×2 = 16
r	2	1010	4	4×2 = 8
i	1	0001	4	1×4 = 4
y	1	10000	5	1×5 = 5
s	2	1011	4	2×4 = 8
n	2	010	3	2×3 = 6
a	2	011	3	2×3 = 6
l	1	10001	5	1×5 = 5
Space	4	001	3	4×3 = 12
.	1	1001	4	1×4 = 4
	25			78 bits



Huffman Tree

A → 65 → 8 bit binary

ASCII

↳ fixed encoding

8 bit / 10 bit

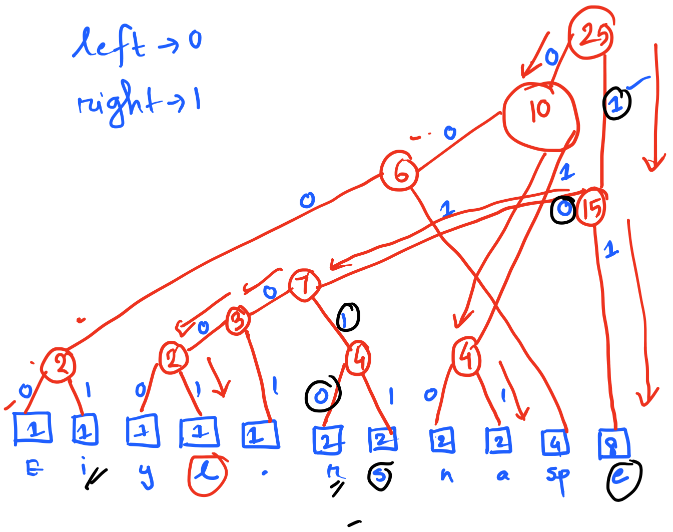
Total bits = (25×8) = 200 bits

Bits saved = (200 − 78) = 122 bits

Huffman Decoding

left → 0

right → 1



25
10
6
15
7
3
2
2
4
4
0
1
1
E
i
y
l
.
r
s
n
a
sp
e

$\underbrace{0000}_{E}\ \underbrace{011}_{a}\ \underbrace{10001}_{l}\ \underbrace{11}_{e}\ \ldots\ldots$